QUANTITATIVE RESEARCH

Independent Research Extensions to a JPMorgan Chase Virtual Experience

Reproducible forecasting, storage valuation, credit-risk modelling and FICO segmentation

Research portfolio edition | Generated from actual repository outputs

Disclaimer: The original virtual-experience assignments are separate from these independently developed research extensions. No JPMorgan Chase endorsement or employment is implied.

Abstract

This report examines four quantitative research tasks using the datasets and experiment outputs included in the accompanying repository. It prioritizes auditable evaluation, clear modelling assumptions and mathematical constraints over claims of production readiness.

Contents

1. Forecasting

2. Storage valuation

3. Credit risk

4. FICO segmentation

5. Reproducibility and limitations

1. Forecasting

Research methodology

Historical mean, seasonal naive, trailing moving average, linear trend, Holt-Winters and SARIMA. Each prediction uses only earlier observations. MAE = mean(|actual - prediction|).

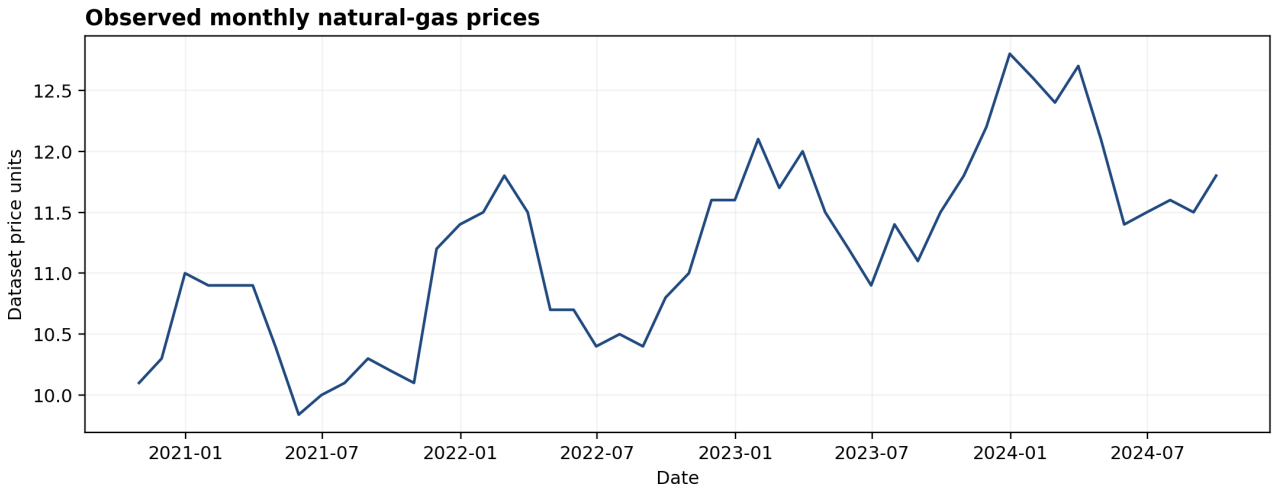
Measured comparison

model	n_test	mae	rmse	mape
holt_winters	24	0.1844	0.2191	1.571
sarima	24	0.2969	0.3587	2.513
moving_average	24	0.4208	0.5095	3.586
linear_trend	24	0.4883	0.5627	4.126
seasonal_naive	24	0.5667	0.6218	4.813
historical_mean	24	0.7951	0.9349	6.611

Interpretation and limitations

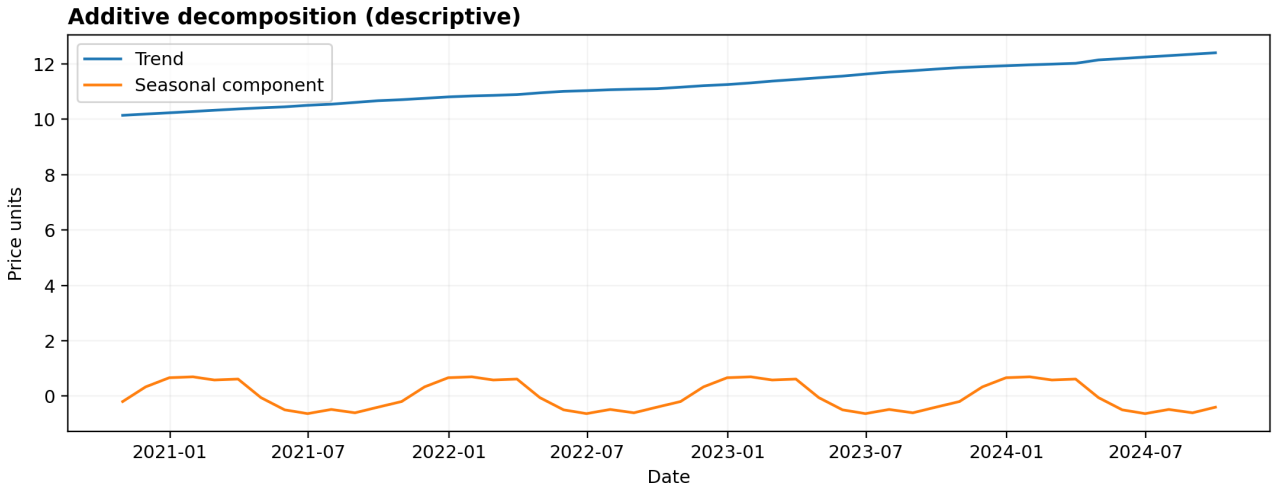
Top by mae: holt_winters, 0.184447. 48 monthly observations; one-step expanding-window test only. Empirical error band uses the same test errors and is not a calibrated prospective interval.

01 History



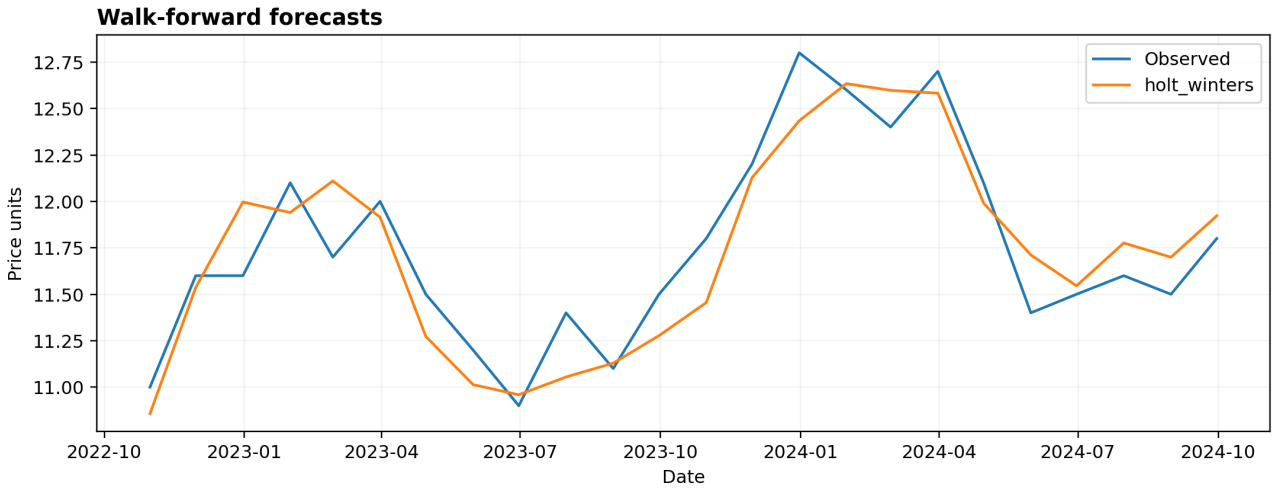
Research visualization: 01 history.

02 Decomposition



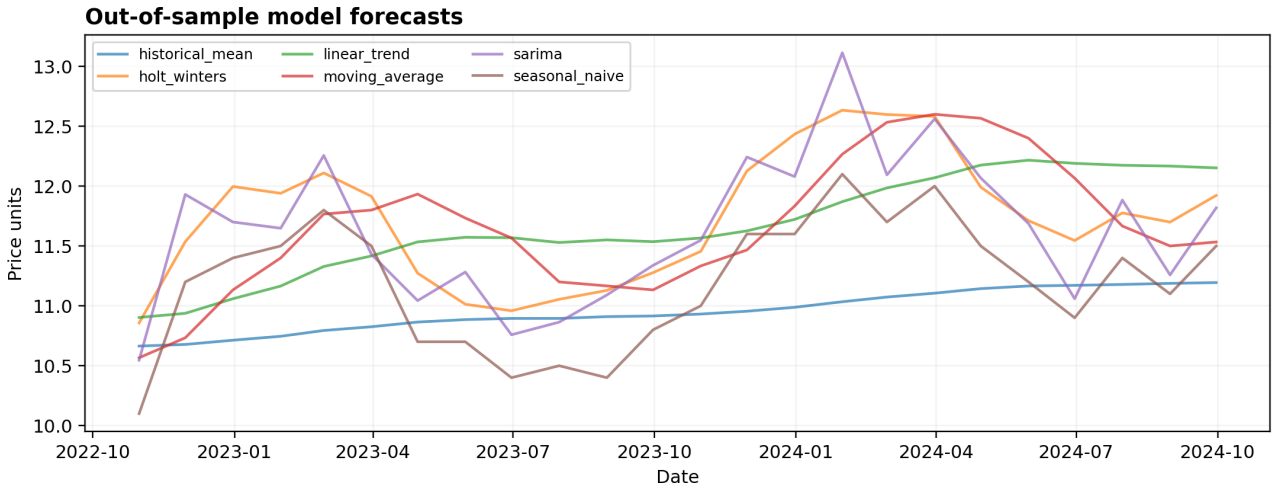
Research visualization: 02 decomposition.

03 Best Model



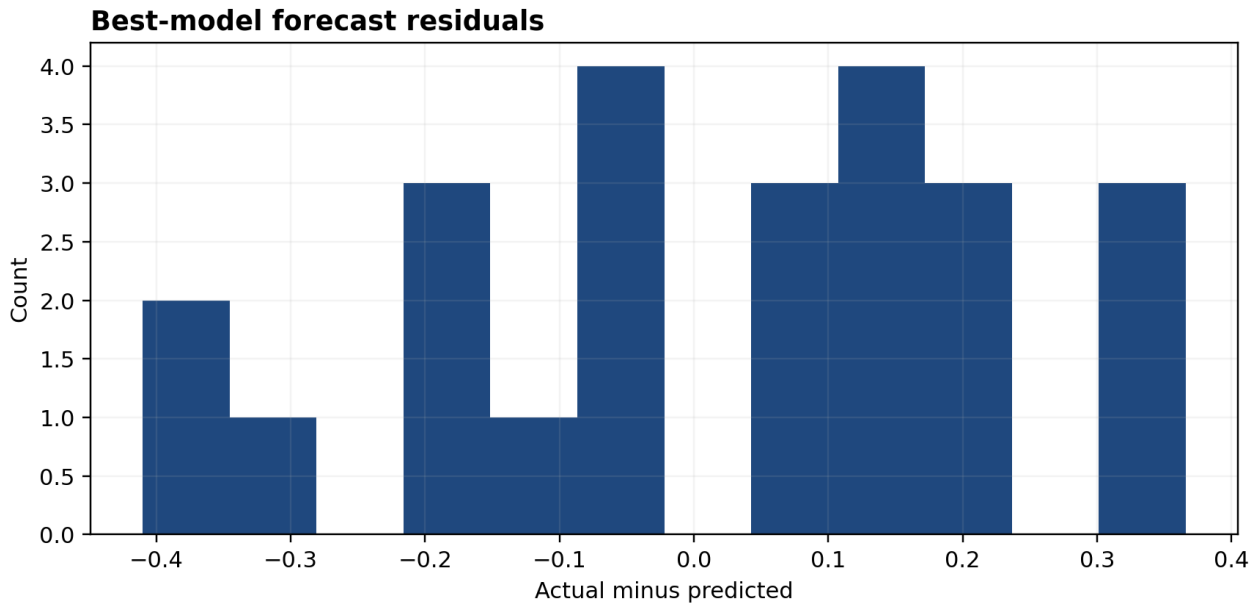
Research visualization: 03 best model.

04 All Models



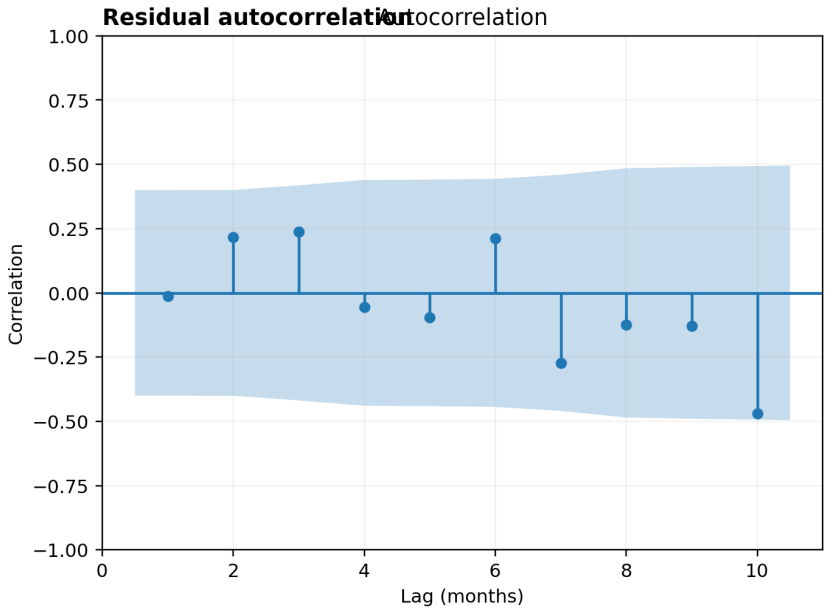
Research visualization: 04 all models.

05 Residual Hist



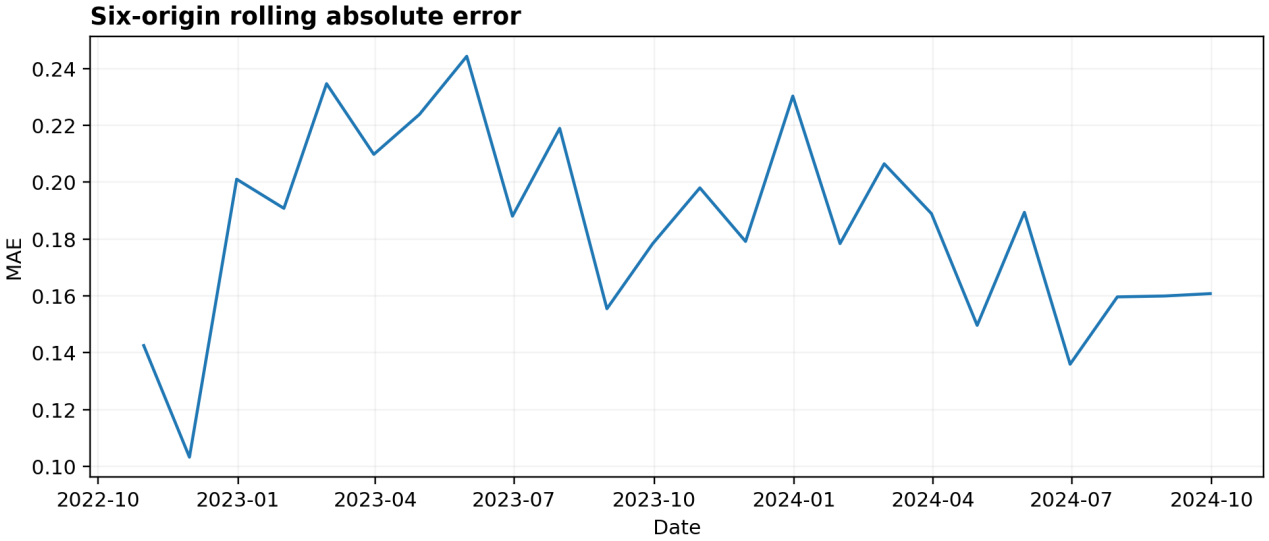
Research visualization: 05 residual hist.

06 Residual Acf



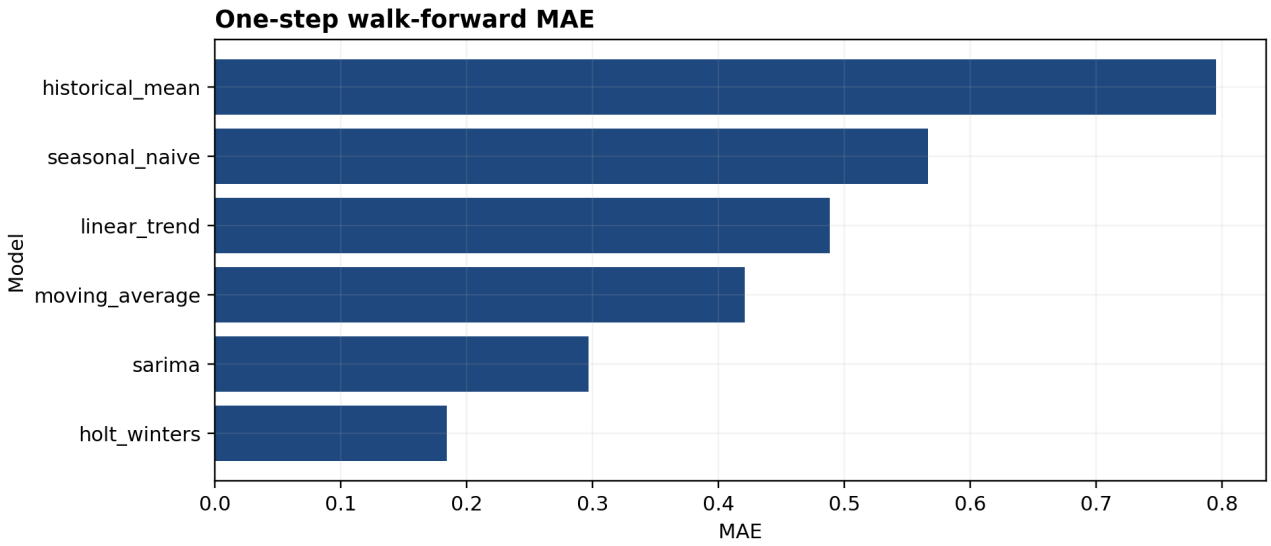
Research visualization: 06 residual acf.

07 Rolling Mae



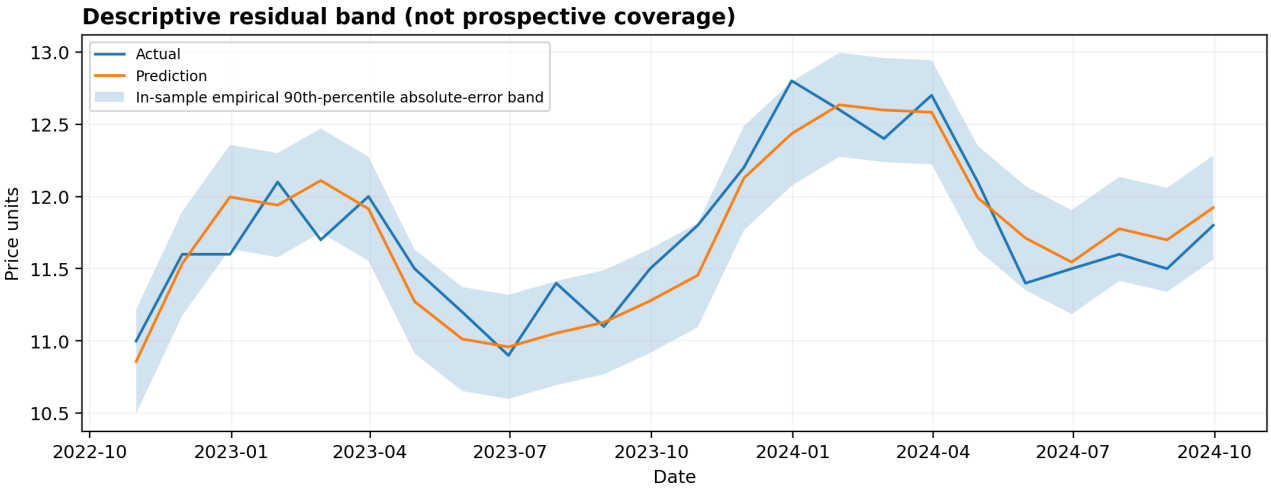
Research visualization: 07 rolling mae.

08 Mae Comparison



Research visualization: 08 mae comparison.

09 Empirical Band



Research visualization: 09 empirical band.

2. Storage valuation

Research methodology

Maximize sum(withdrawals * (price - transaction fee) - injections * (price + transaction fee) - storage fee), subject to inventory conservation, capacity, flow limits and terminal inventory.

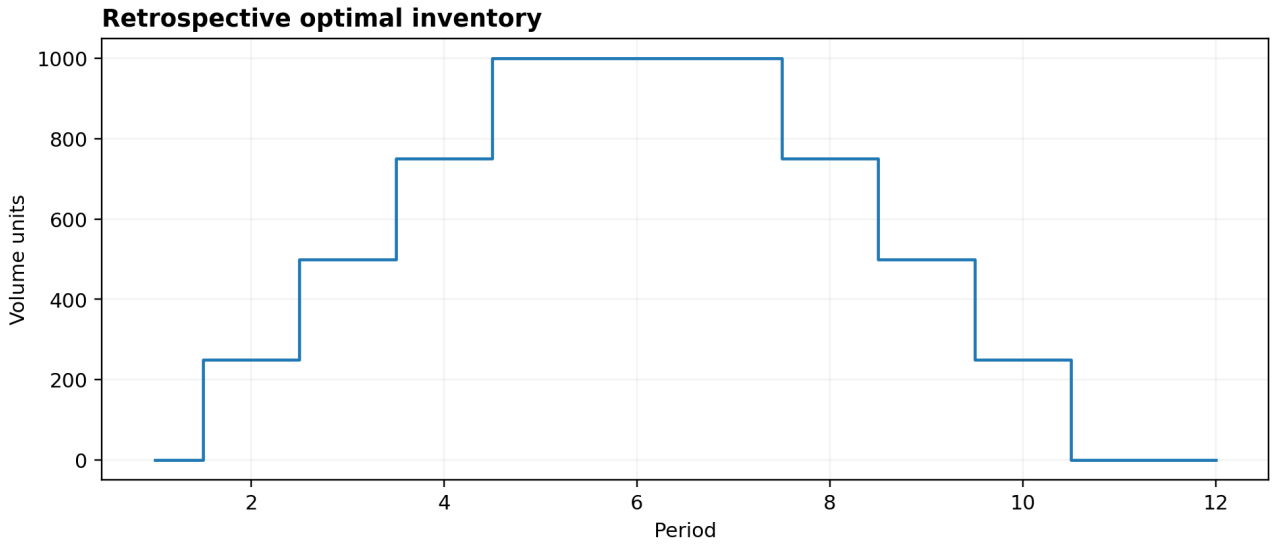
Measured comparison

scenario	npv	injection_cost	withdrawal_revenue	storage_cost
base_assumption	1231	2460	3715	24
low_price_assumption	976	1970	2970	24
high_price_assumption	1486	2950	4460	24
higher_storage_fee	1075	2460	3715	180
reduced_capacity	741	1180	1945	24
higher_transaction_fee	951	2600	3575	24
restricted_withdrawal	687	1682	2393	24

Interpretation and limitations

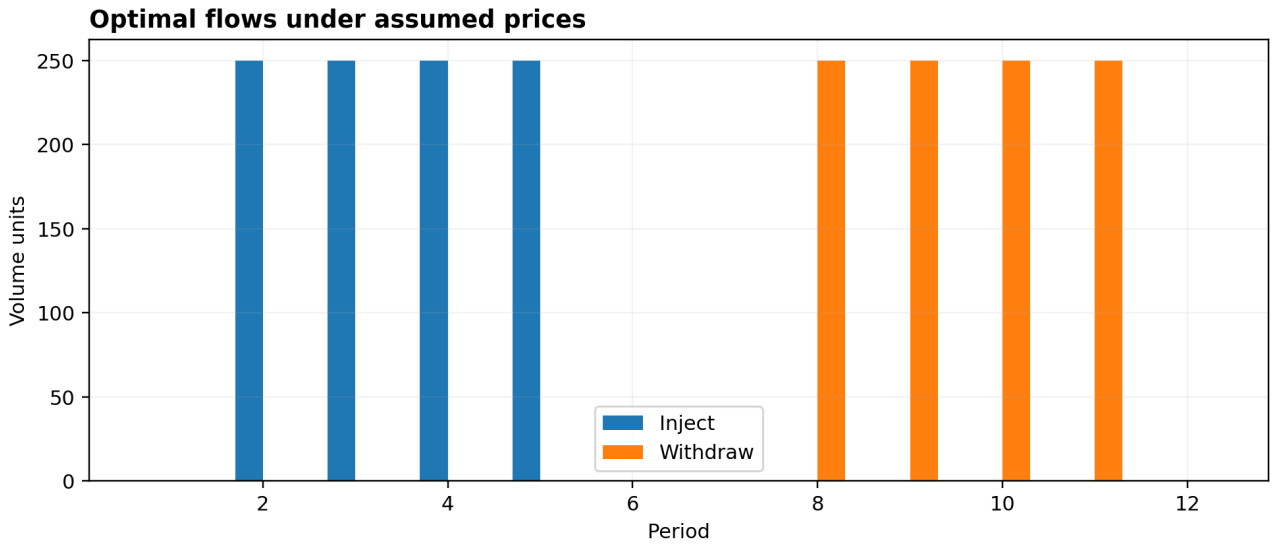
Top by npv: high_price_assumption, 1486. All prices are illustrative assumptions. LP uses perfect foresight and the displayed NPV is undiscounted (zero discount rate). No implementable real-time dispatch is established.

01 Inventory



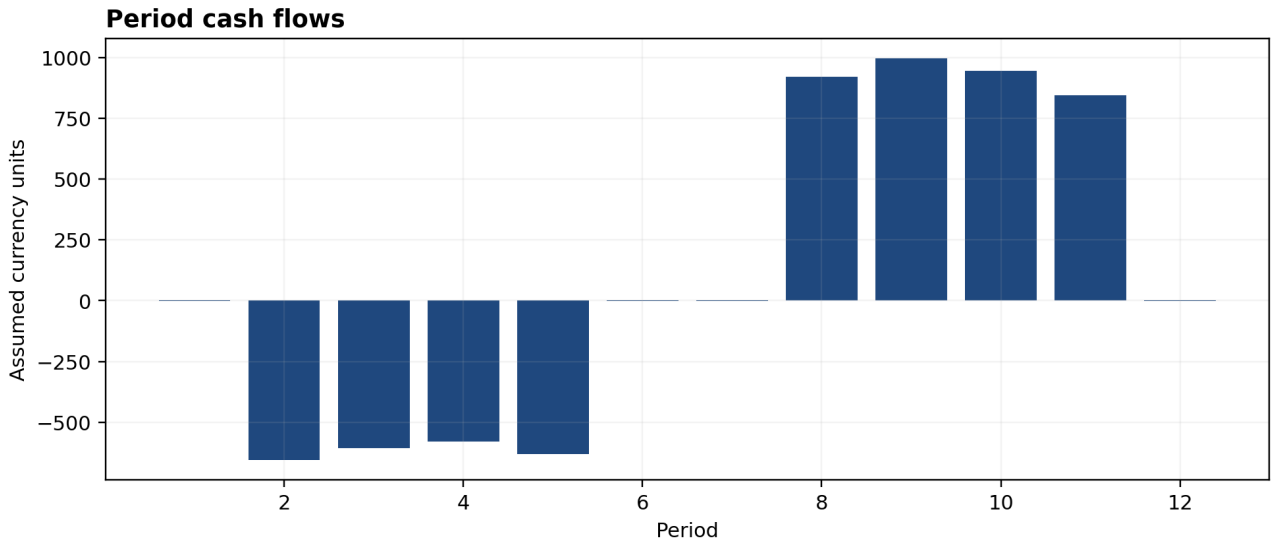
Research visualization: 01 inventory.

02 Flows



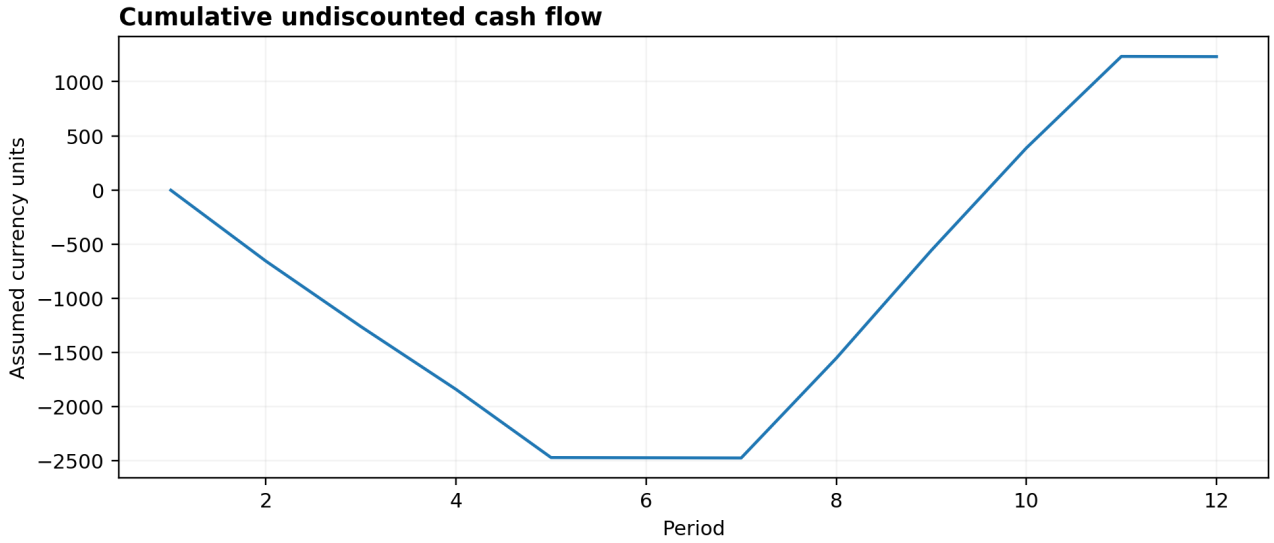
Research visualization: 02 flows.

03 Cashflows



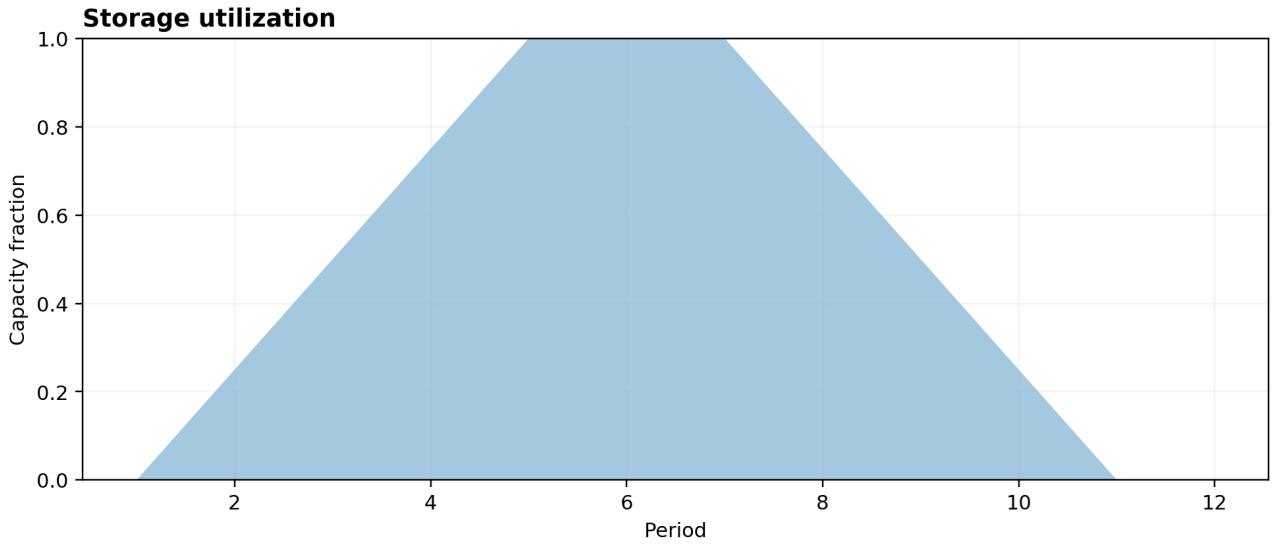
Research visualization: 03 cashflows.

04 Cumulative



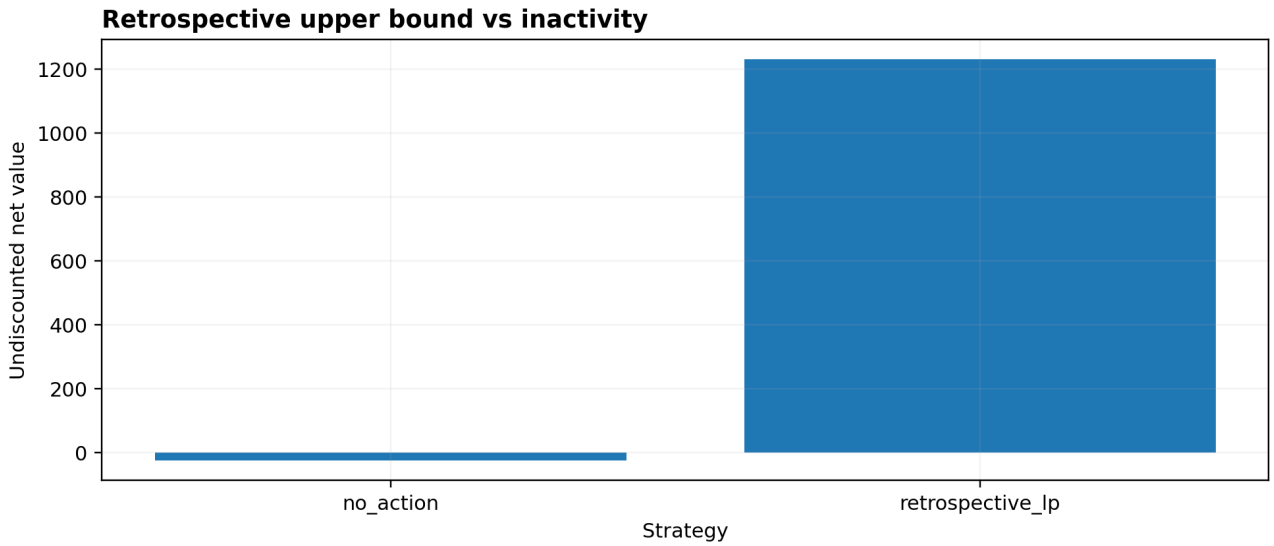
Research visualization: 04 cumulative.

05 Utilization



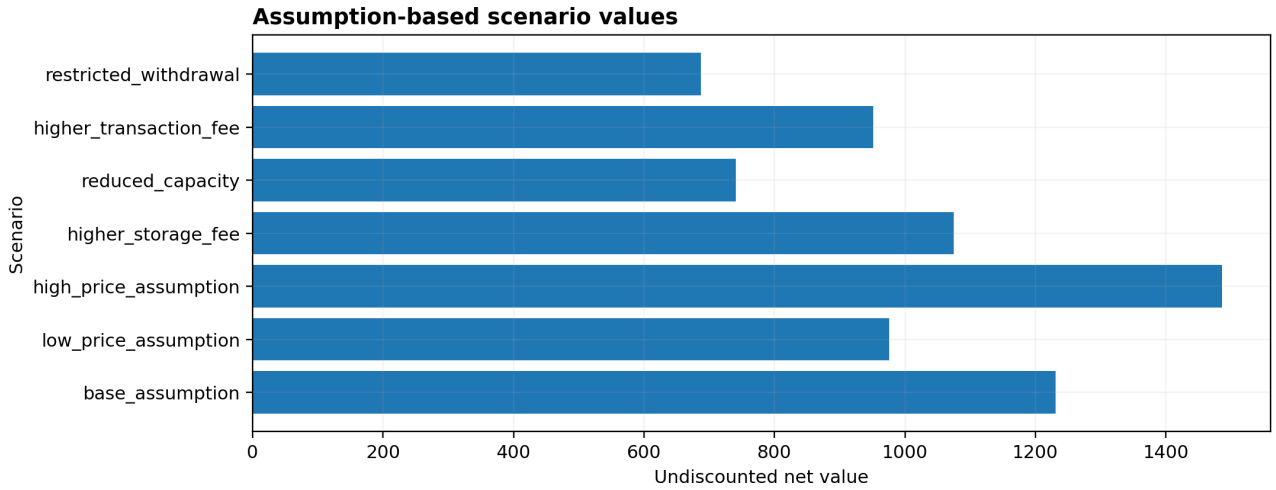
Research visualization: 05 utilization.

06 Strategy



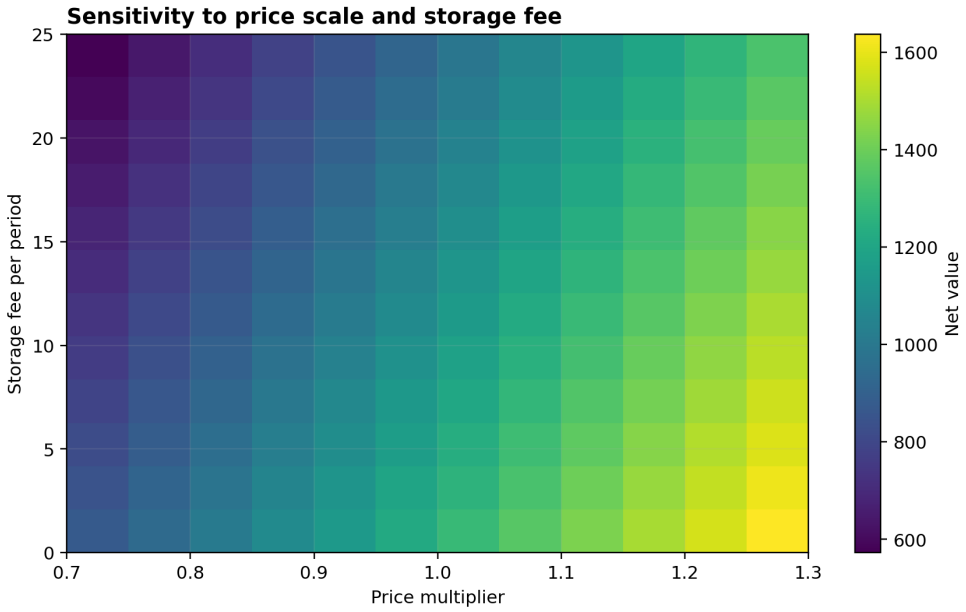
Research visualization: 06 strategy.

07 Scenarios



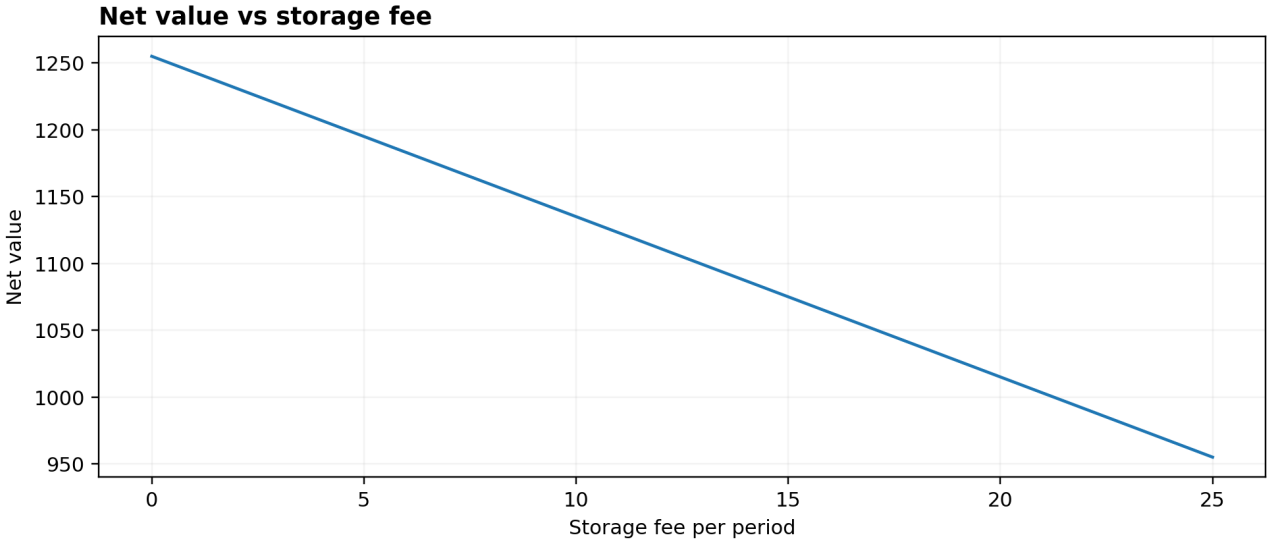
Research visualization: 07 scenarios.

08 Sensitivity



Research visualization: 08 sensitivity.

09 Storage Fee



Research visualization: 09 storage fee.

3. Credit risk

Research methodology

Stratified 75/25 split; preprocessing fit within training pipeline. ROC-AUC and PR-AUC are discrimination metrics; Brier and log loss measure probabilistic quality. Expected loss = PD x LGD x EAD.

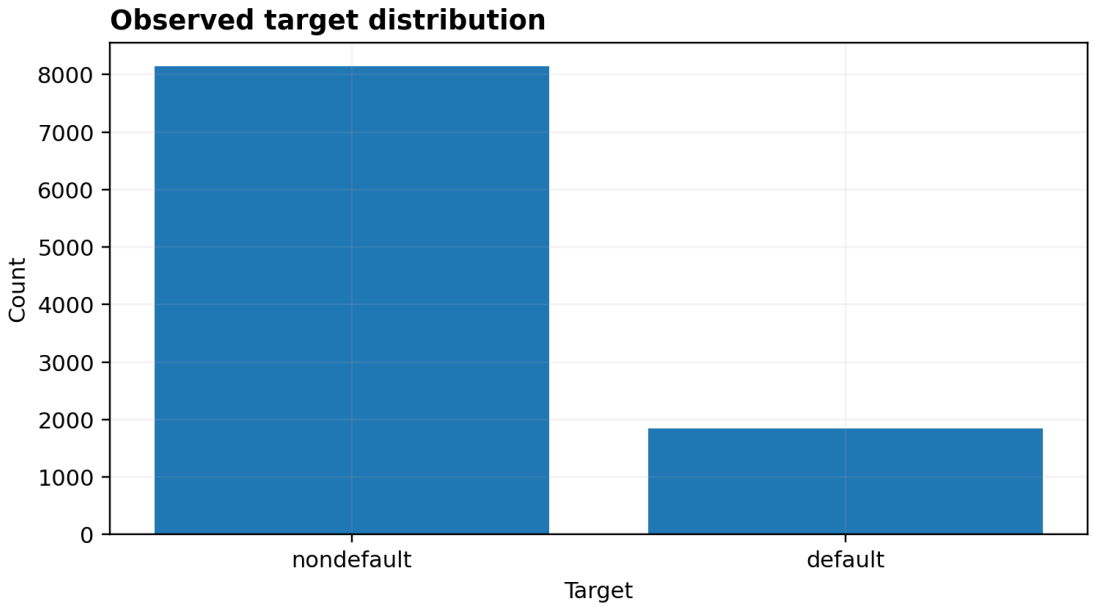
Measured comparison

model	roc_auc	pr_auc	precision	recall
logistic_regression	1	1	1	0.9914
hist_gradient_boosting	0.9999	0.9997	0.9892	0.9914
random_forest	0.9998	0.9991	0.9891	0.9827
decision_tree	0.9977	0.9863	0.9725	0.9179

Interpretation and limitations

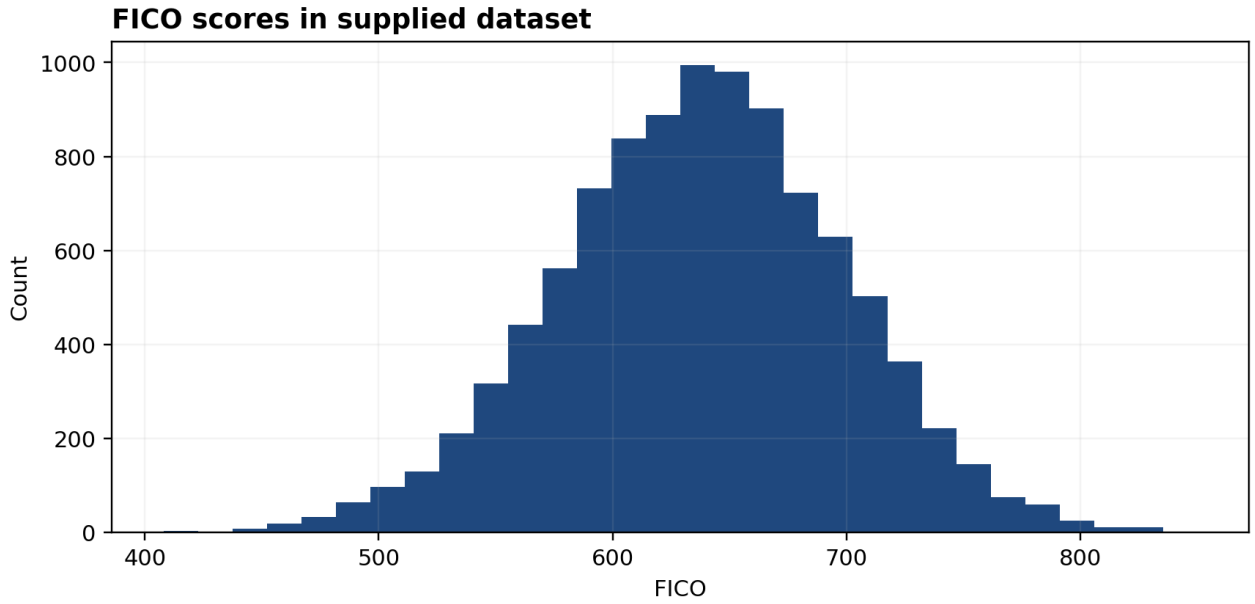
Top by roc_auc: logistic_regression, 0.999998. The data provenance is not verified. Near-perfect discrimination may reflect target construction; temporal and external validation are unavailable. LGD is assumed, not observed.

01 Class Balance



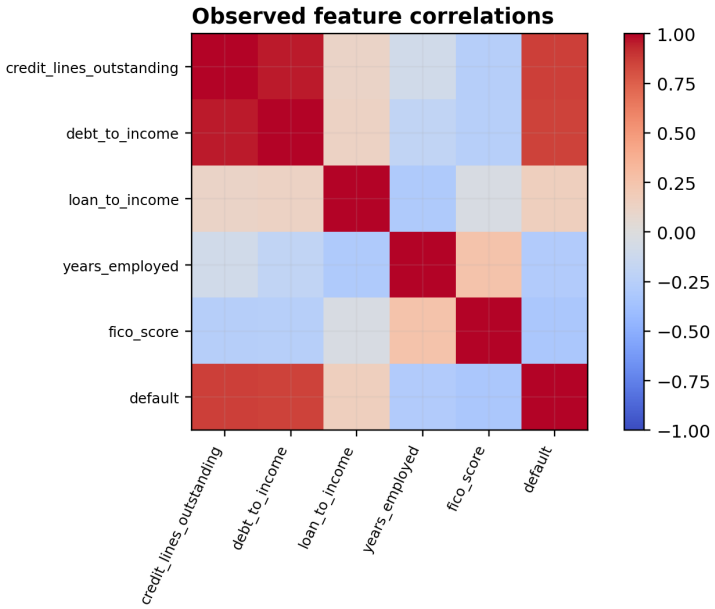
Research visualization: 01 class balance.

02 Fico Distribution



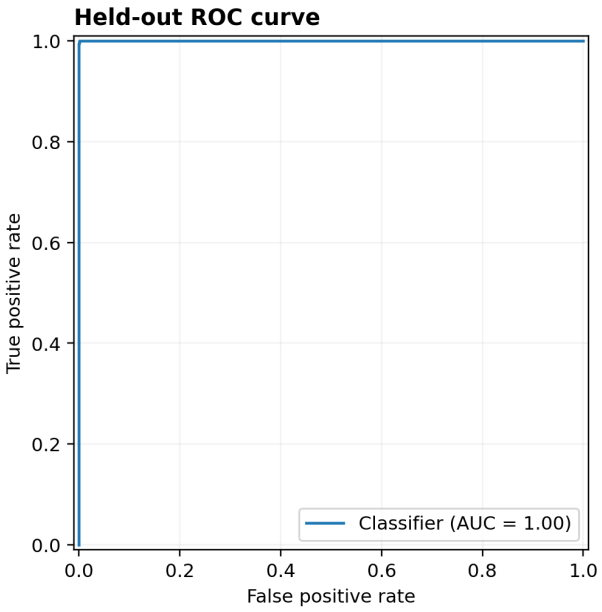
Research visualization: 02 fico distribution.

03 Correlation



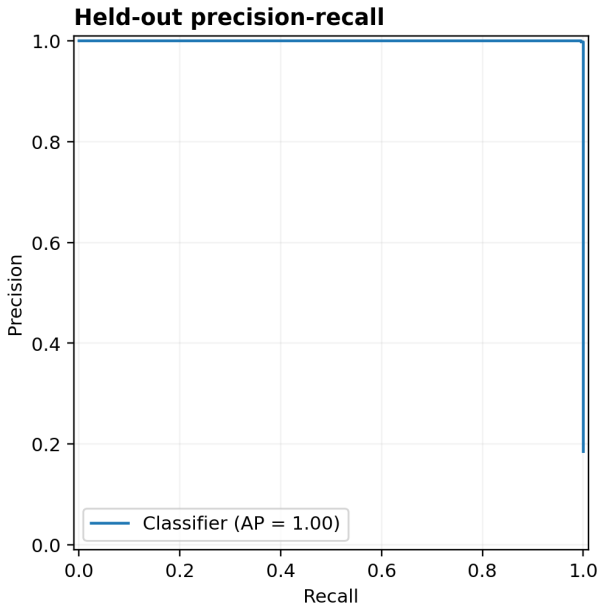
Research visualization: 03 correlation.

04 Roc



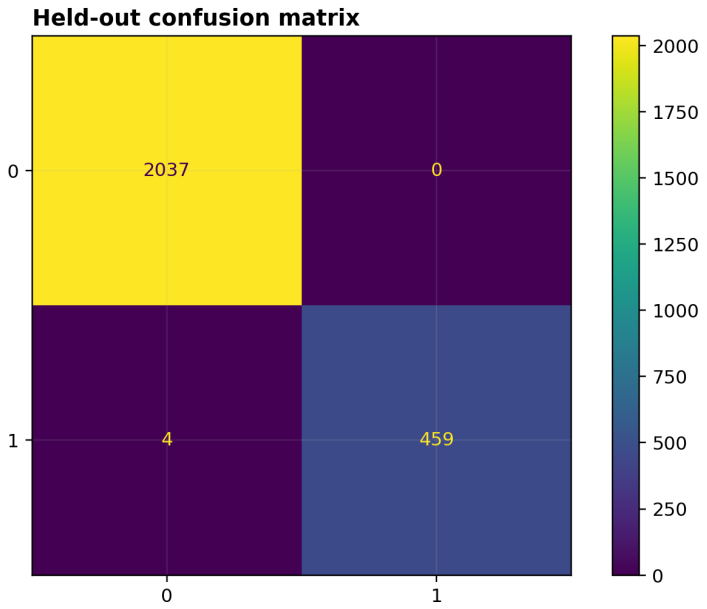
Research visualization: 04 roc.

05 Pr



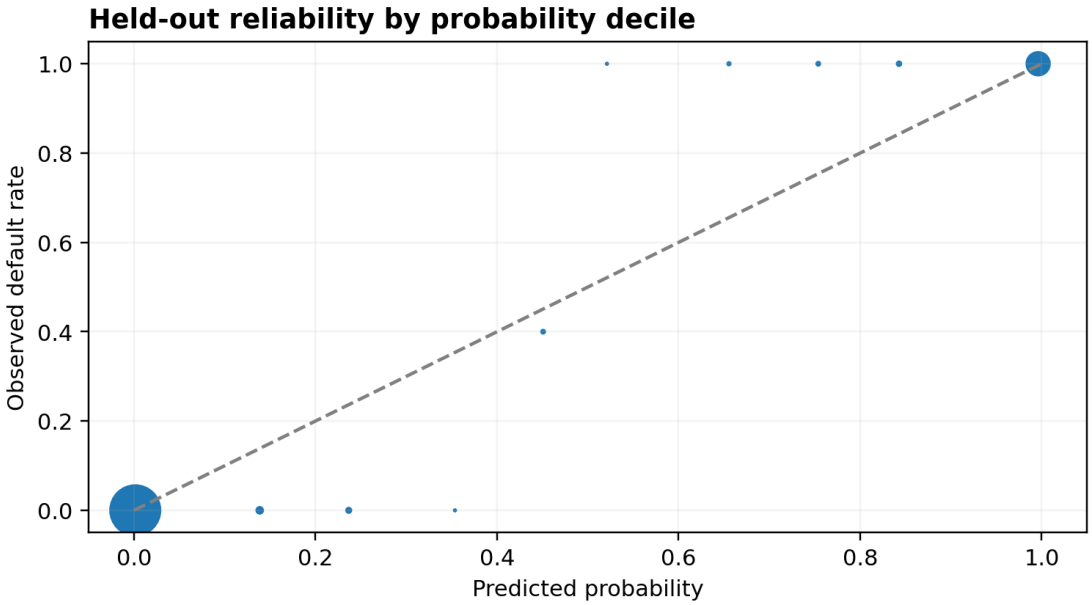
Research visualization: 05 pr.

06 Confusion



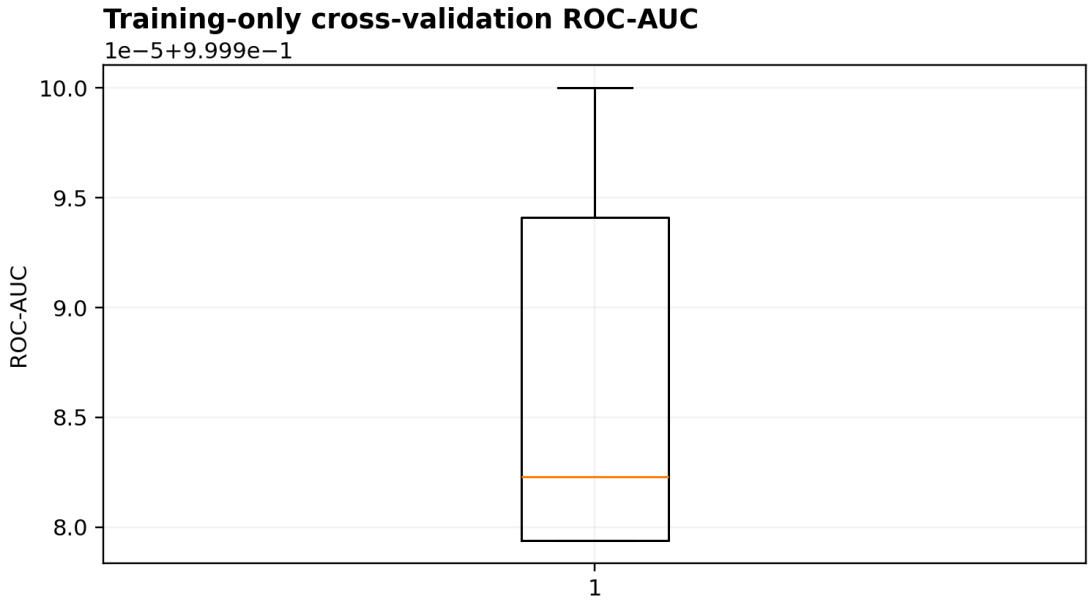
Research visualization: 06 confusion.

07 Calibration



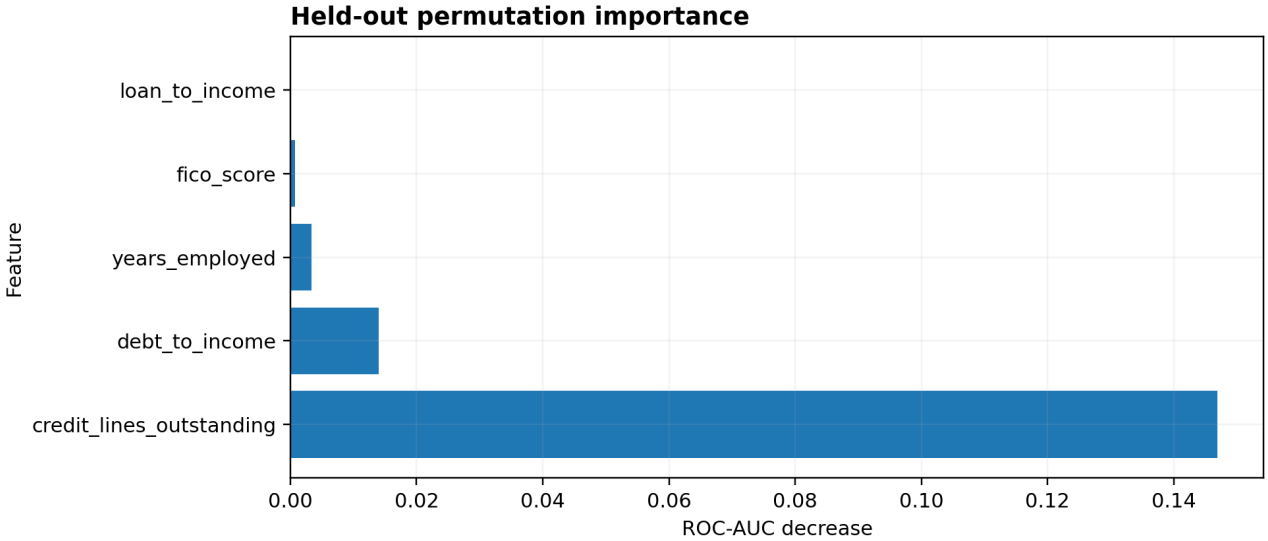
Research visualization: 07 calibration.

08 Cv



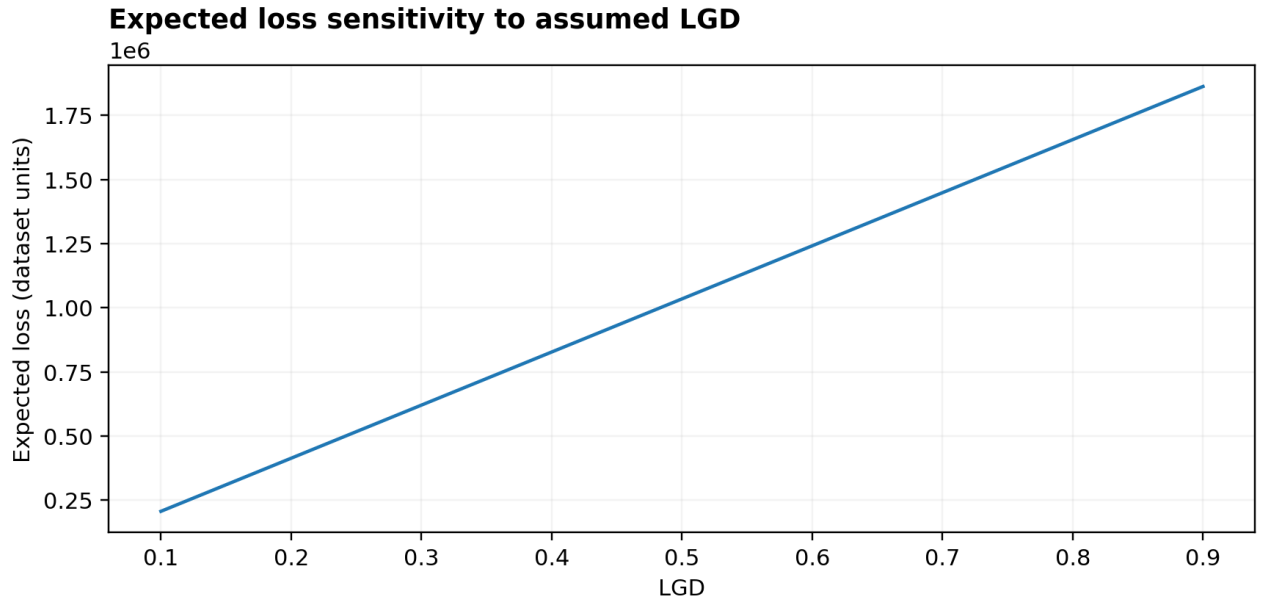
Research visualization: 08 cv.

09 Permutation



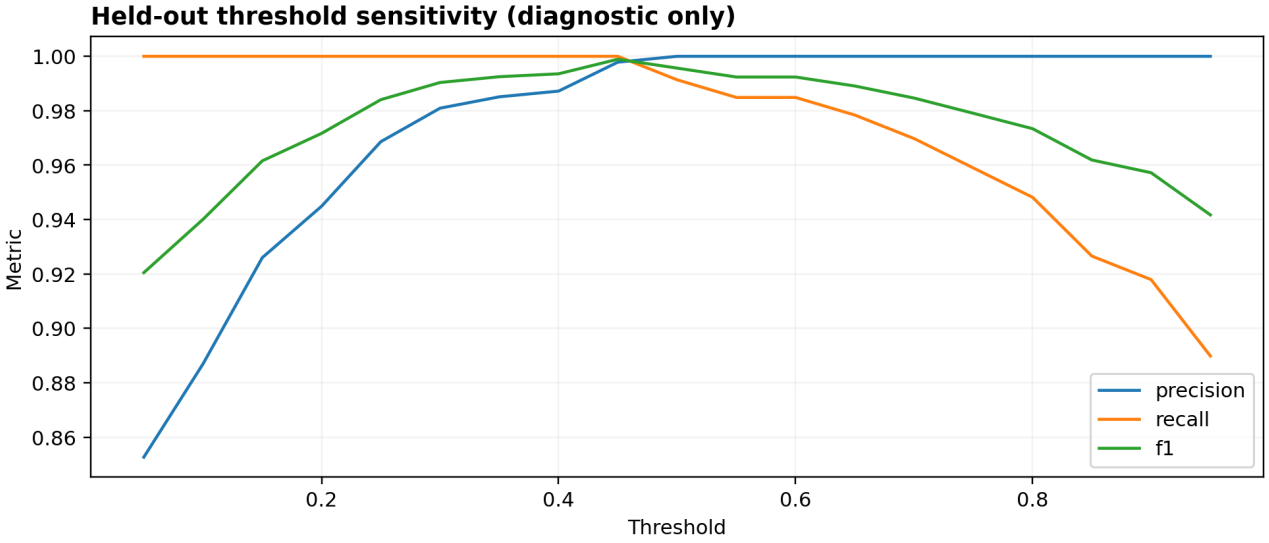
Research visualization: 09 permutation.

10 Expected Loss



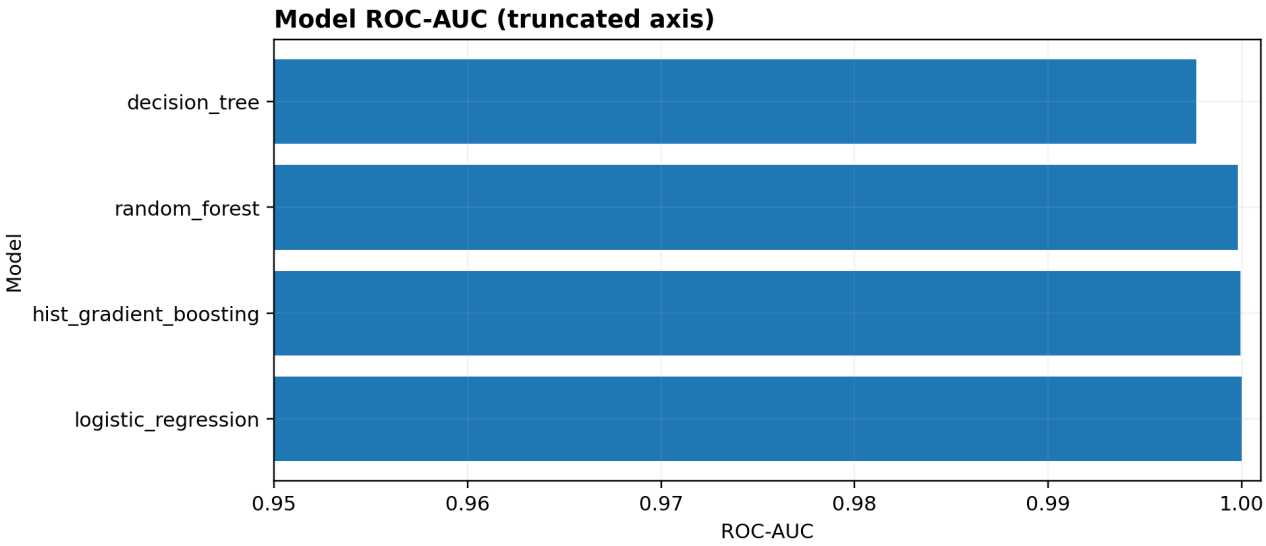
Research visualization: 10 expected loss.

11 Thresholds



Research visualization: 11 thresholds.

12 Comparison



Research visualization: 12 comparison.

4. FICO segmentation

Research methodology

Bucket boundaries are fit using training scores only. Held-out log-likelihood sums y log(p) + (1-y) log(1-p). WoE = log(nondefault share / default share), with 0.5 event smoothing.

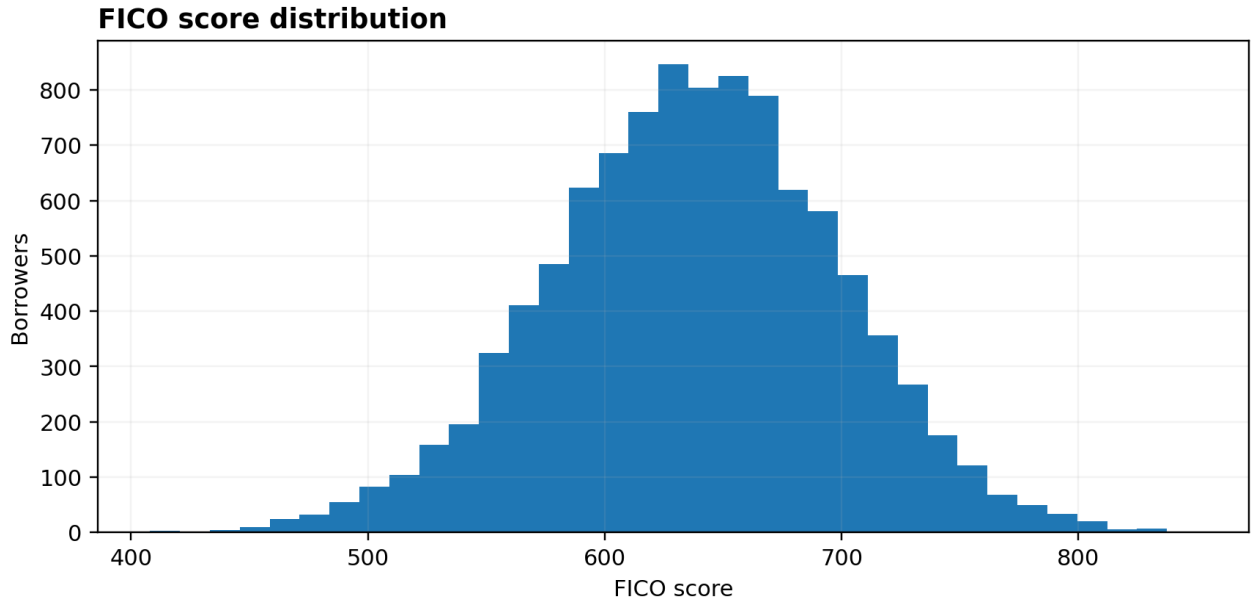
Measured comparison

method	buckets	test_log_likelihood	min_train_bucket	monotonic_train
dynamic_programming	10	-1086	96	0
equal_width	10	-1091	1	1
equal_frequency	10	-1099	712	1
quantile	10	-1099	712	1

Interpretation and limitations

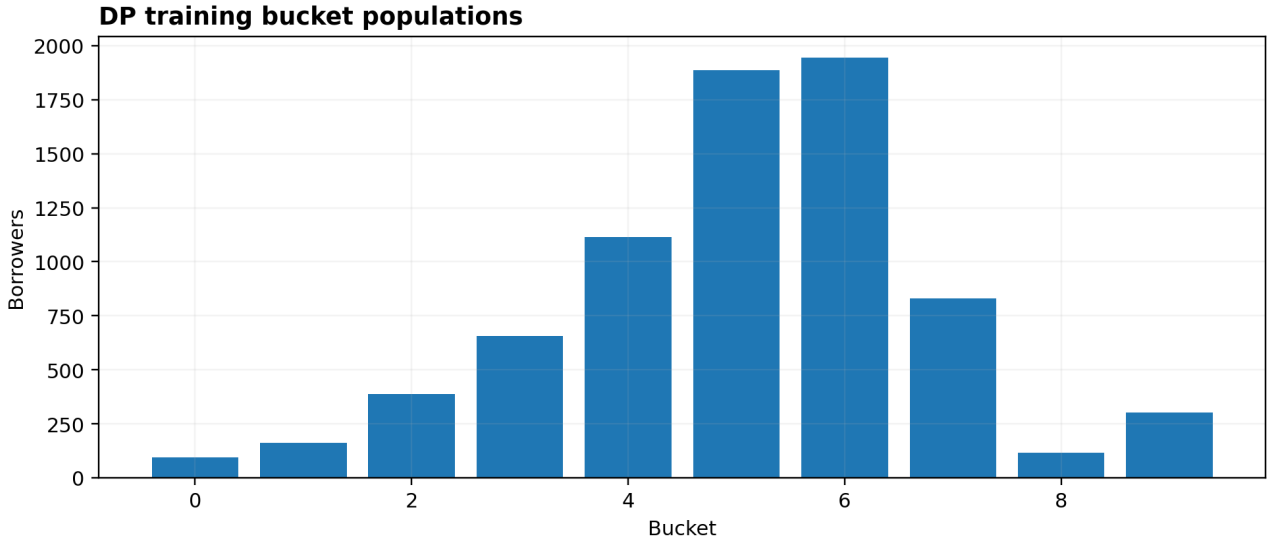
Top by test_log_likelihood: dynamic_programming, -1086.07. Train-fitted buckets are evaluated on held-out borrowers. Dynamic programming can overfit and the reported WoE/IV are train-only. Bootstrap stability remains untested.

01 Score Distribution



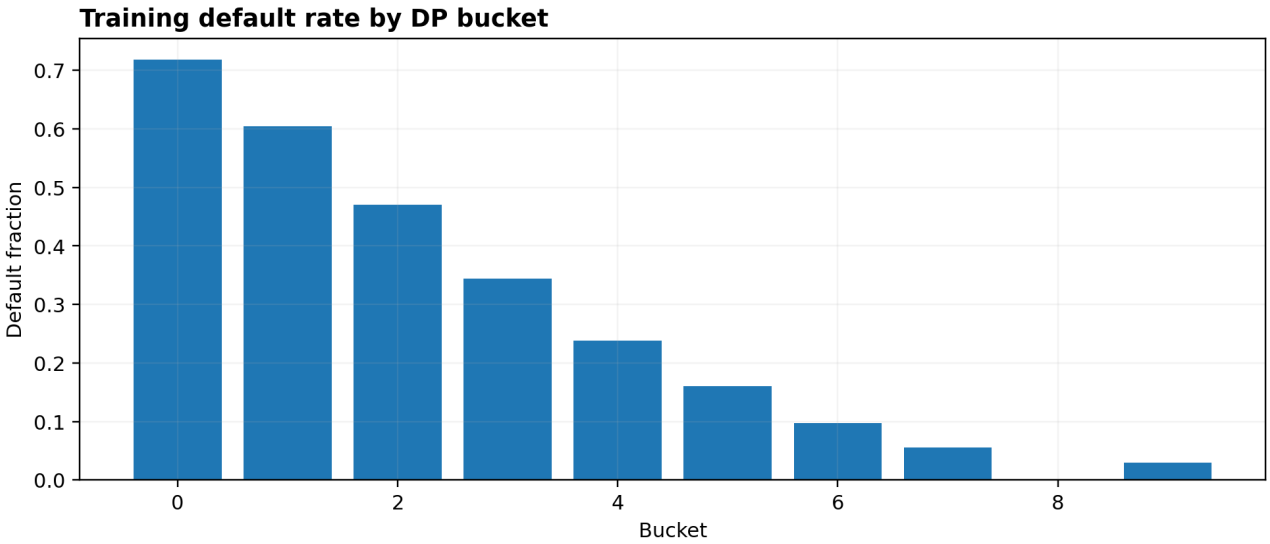
Research visualization: 01 score distribution.

02 Population



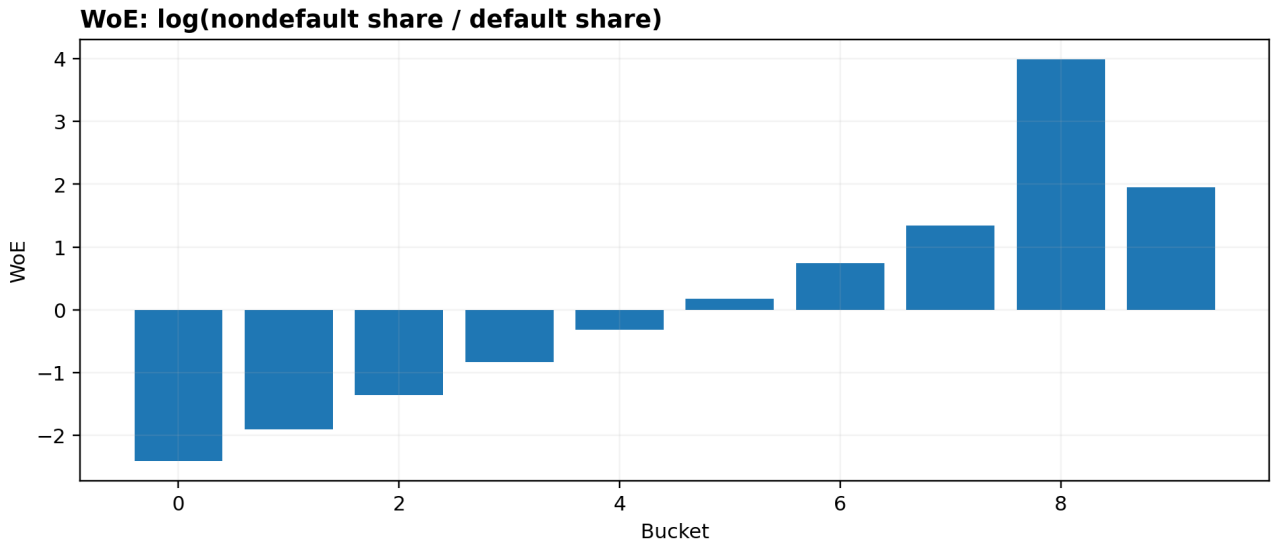
Research visualization: 02 population.

03 Default Rate



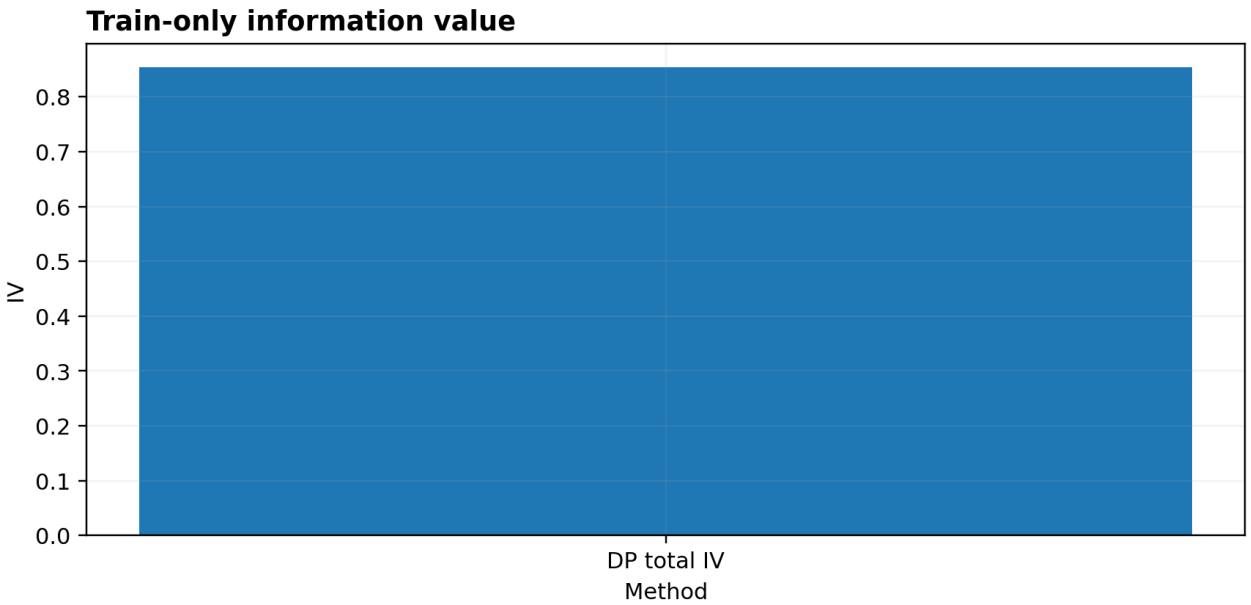
Research visualization: 03 default rate.

04 Woe



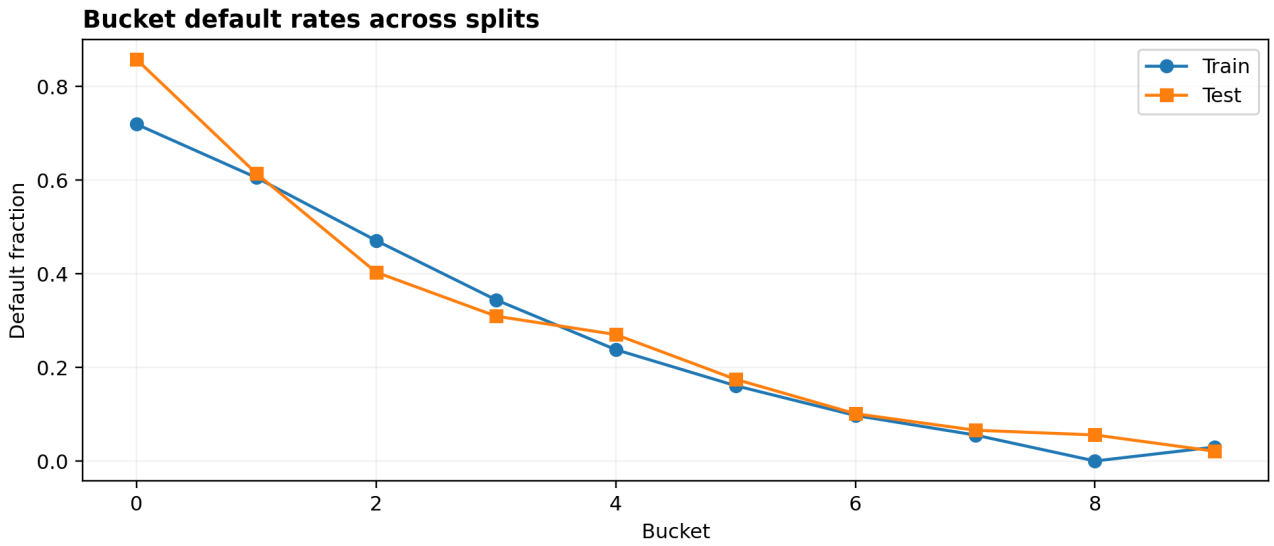
Research visualization: 04 woe.

05 Iv



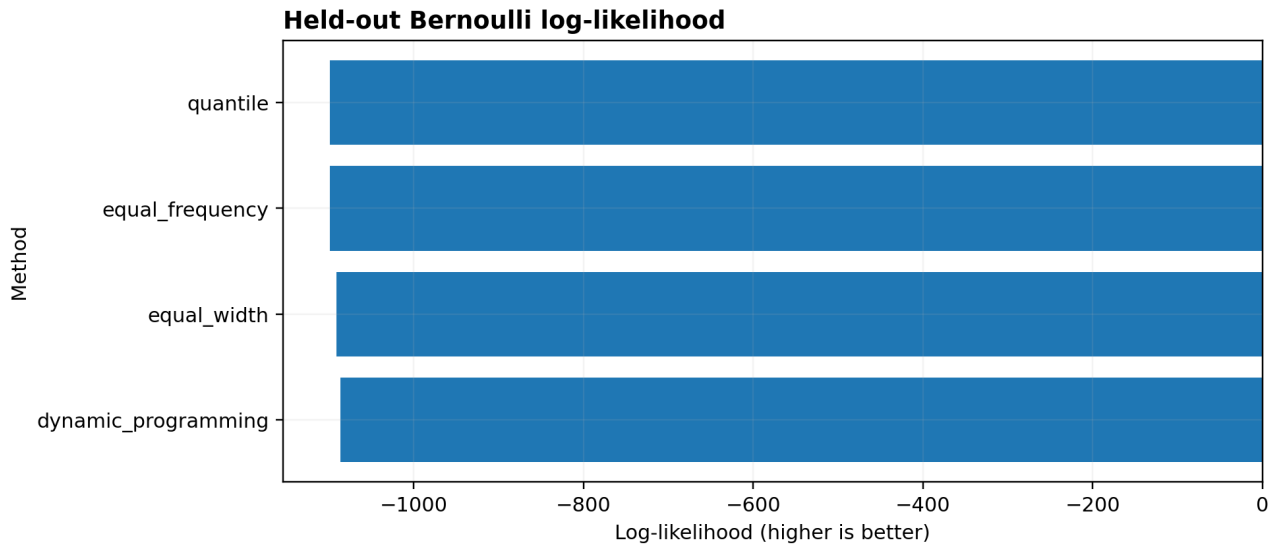
Research visualization: 05 iv.

06 Train Test



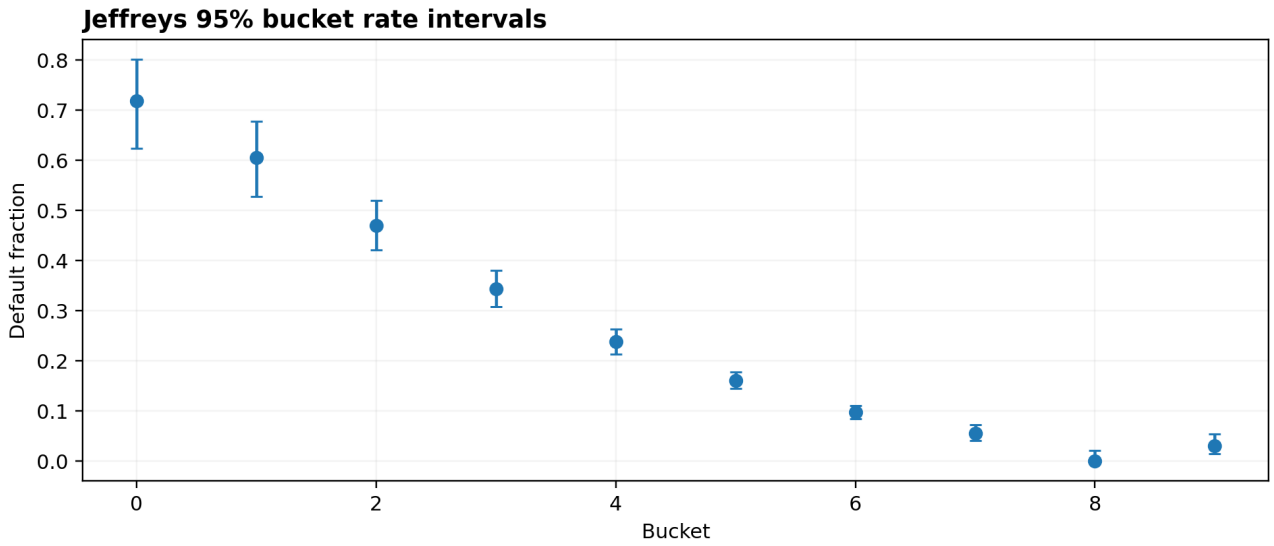
Research visualization: 06 train test.

07 Method Comparison



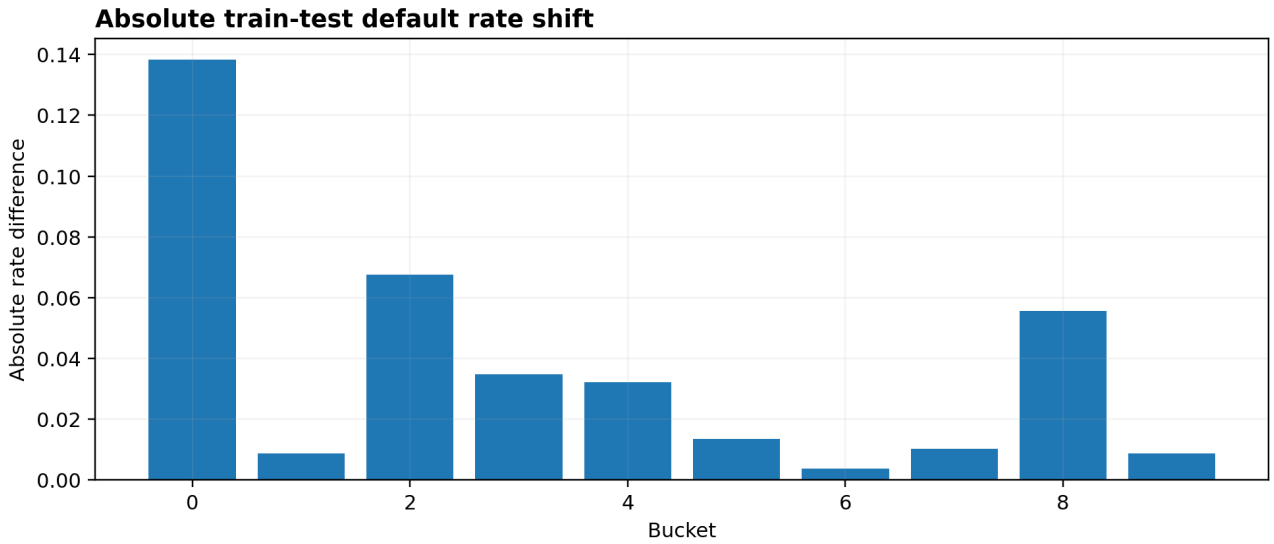
Research visualization: 07 method comparison.

08 Bucket Intervals



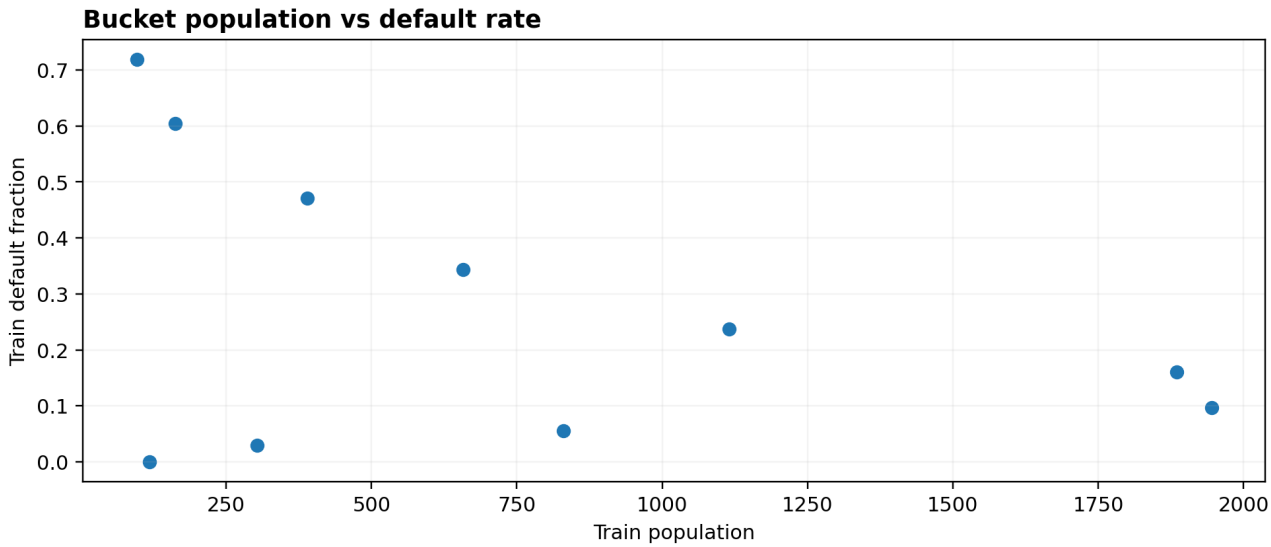
Research visualization: 08 bucket intervals.

09 Stability



Research visualization: 09 stability.

10 Population Vs Risk



Research visualization: 10 population vs risk.

5. Reproducibility and limitations

Run pip install -r requirements.txt, then python -m scripts.run_all_experiments, python -m scripts.generate_all_figures, python -m scripts.generate_all_reports, and python -m pytest -q. Dashboard requires pip install -r requirements-dashboard.txt and streamlit run dashboard/app.py.

Limitations

48 monthly observations; one-step expanding-window test only. Empirical error band uses the same test errors and is not a calibrated prospective interval.

All prices are illustrative assumptions. LP uses perfect foresight and the displayed NPV is undiscounted (zero discount rate). No implementable real-time dispatch is established.

The data provenance is not verified. Near-perfect discrimination may reflect target construction; temporal and external validation are unavailable. LGD is assumed, not observed.

Train-fitted buckets are evaluated on held-out borrowers. Dynamic programming can overfit and the reported WoE/IV are train-only. Bootstrap stability remains untested.

References and methods

Hyndman and Athanasopoulos, Forecasting: Principles and Practice; scikit-learn model evaluation documentation; statsmodels time-series documentation; SciPy HiGHS linear programming documentation. This report does not claim peer review.